Lords Institute of Engineering & Technology

Department of Computer Science and Engineering

MAJOR-PROJECT PRESENTATION - II

SecureAuthX: A Cross-Browser Extension for Secure Password Management and Website Security Analysis

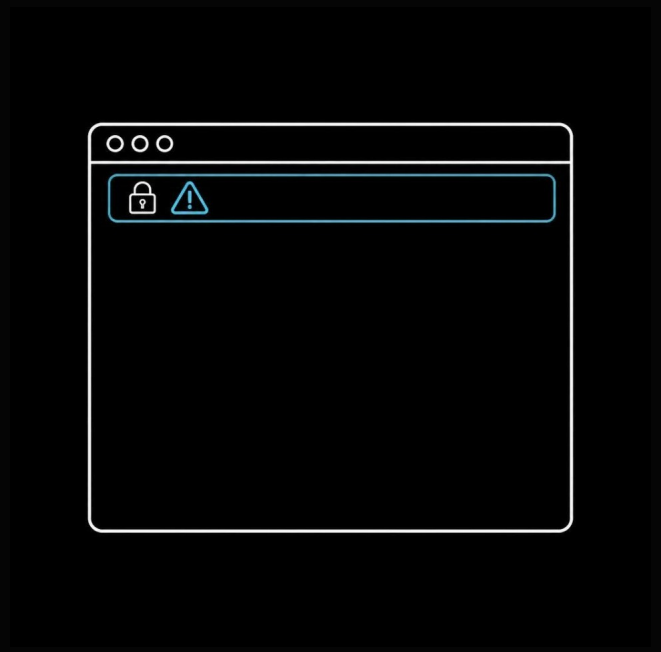
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PROBLEM AND PROPOSED SOLUTION

SecureAuthX connects password protection with website security context within a single browser extension.



First-review problem

Weak or reused passwords, phishing and unsafe destinations expose users to risk. Security and privacy tools are fragmented.

Proposed solution

A local vault, password analysis, website and privacy signals, and risk assessment in one extension.

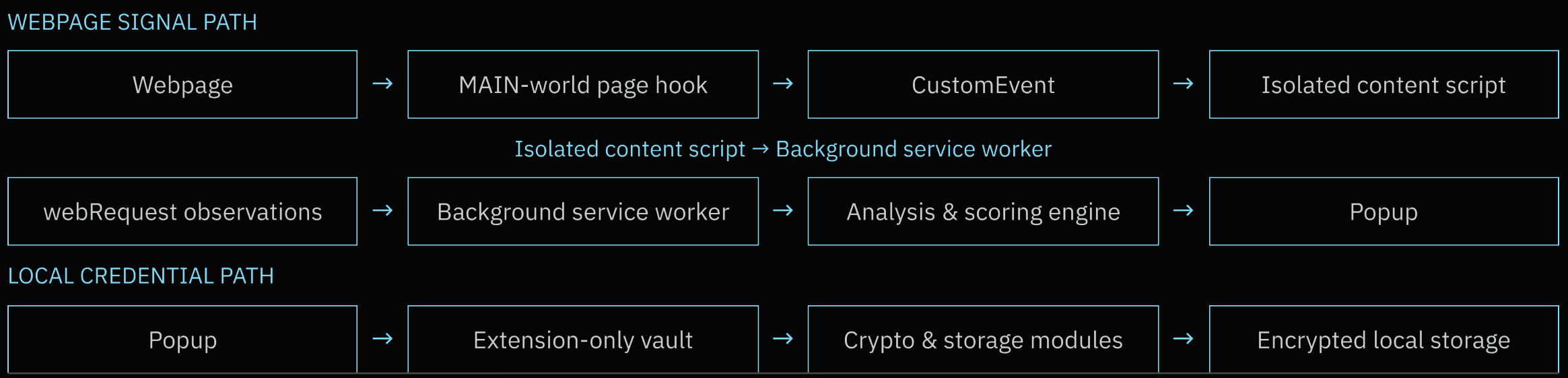
IMPLEMENTATION STATUS

From proposal to implementation

Module	Status
Password Derivation	● Implemented
Encrypted Local Vault	● Implemented
Password Strength and Reuse Analysis	● Implemented
Secure Autofill	● Implemented in code; Phase 3C validation pending
Website Security Analysis	● Implemented with bounded detection
Privacy and Fingerprinting Analysis	◐ Partial coverage
Cross-Browser Validation	○ Pending; browsers beyond Chromium unverified

SYSTEM ARCHITECTURE

Architecture connects page signals to local credential actions



User-initiated Fill returns through the top-frame content script to the webpage, after checks.

Master password and decrypted vault remain outside persistent storage.

TECHNOLOGY STACK

Implementation stack is Chromium-oriented

Extension

HTML, CSS, JavaScript ES modules, Manifest V3 and browser APIs.

Security

Web Crypto API, PBKDF2-HMAC-SHA-256, AES-256-GCM and browser local storage.

Testing

Node.js built-in test runner and Playwright 1.56.0 for unit and E2E tests.

Browsers beyond Chromium remain unverified.

LOCAL CREDENTIAL PROTECTION

Vault and autofill protect credentials locally

Password derivation

Master password + website domain + account identity → PBKDF2-SHA-256 → derived site-specific password.

250,000

PBKDF2 iterations · 256-bit derived material

Vault encryption

PBKDF2-derived vault key → AES-256-GCM encryption → ciphertext-only local persistence.

Random salt; fresh 96-bit IV per save.

Explicit Fill checks origin, exact hostname, HTTPS or loopback, and lookalike conditions. Lock manually, after inactivity or worker stop. Phase 3C test cases exist; results are not documented.

WEBSITE AND PRIVACY ANALYSIS

Security and privacy analysis combines bounded signals

Website security

- HTTPS
- Form destinations
- Lookalike hostnames
- Security headers and cookie flags

Fingerprinting

- Selected canvas, WebGL and audio signals
- Timezone, locale and font signals

Tracker detection

- Local static signatures

Scoring and limits

Critical-risk findings cap the score. Detection is rule-based: not comprehensive phishing detection or exhaustive fingerprinting coverage.

TEST EVIDENCE AND LIMITATIONS

Phase 2 passed 213 documented tests; Phase 3 awaits execution evidence

213

Documented Phase 2 passes

150 unit tests and 63 Playwright E2E tests passed.

Five defect-targeting E2E tests failed before fixes and passed after.

Phase 3

Phase 3C adversarial test files are present, but executed results are not yet documented in the supplied report.

Real 30-minute wall-clock timeout untested (timer tested deterministically). OS clipboard and browsers beyond Chromium untested. JavaScript memory zeroing not guaranteed; exact hostname matching does not distinguish ports.

NEXT REVIEW EVIDENCE

Next: verify Phase 3 and complete review evidence

SecureAuthX has progressed from a proposed security extension to an implemented local-first credential protection and website analysis system. Further validation and cross-browser testing remain necessary before final release.

Execute and capture
Execute and capture Phase 3 unit and E2E results.

Address limitations
Address limitations and validate additional browsers.

